STA380 Project Proposal

Permutation Test on ECG-Derived "Age Gap" Using the CODE-15% Dataset (Zenodo 4916206)

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We propose a project under **Theme 3, Topic 11: Permutation Test**. We will use the publicly available **CODE-15% ECG dataset** hosted on Zenodo (record **4916206**) [Ribeiro et al., 2021]. The app will implement a two-sample permutation test comparing subjects with atrial fibrillation (AF) versus those without AF.

1 Project Topic

Permutation Test (Advanced +2): a two-sample permutation test for detecting differences between **AF** and **non-AF** groups on an ECG-derived continuous outcome.

2 Simulation vs. Dataset

This project is **dataset-based**. We will extract relevant variables from the dataset metadata file (e.g., `exams.csv`), including:

- `AF` (binary group label),
- `age` (chronological age),
- `nn_predicted_age` (neural-network predicted "ECG age").

Our main analysis does not require raw waveform files, but if time permits we will include an optional extension using simple waveform features as an additional outcome.

3 Project Details

Research question. Do patients with **AF** and **non-AF** have statistically significant differences in an ECG-derived "age gap"?

Groups (two samples).

- Group 1: `AF == TRUE` (AF)
- Group 2: `AF == FALSE` (non-AF)

Outcome variable. We define the continuous outcome ("Age Gap"):

$$g = \texttt{nn\_predicted\_age} - \texttt{age}.$$

Intuitively, $g > 0$ means the ECG-based model predicts an "older" physiological age than chronological age, while $g < 0$ suggests the opposite.

Hypotheses.

- H_0: The distribution of g is the same for AF and non-AF (group label is exchangeable).
- H_1: The distributions differ (mean/median/overall shape differs).

Test statistic (user-selectable). To demonstrate flexibility of permutation testing, users can choose:

1. Mean difference: $T = \bar{g}_{\text{AF}} - \bar{g}_{\text{nonAF}}$

2. Median difference: $T = \mathrm{med}(g_{\mathrm{AF}}) - \mathrm{med}(g_{\mathrm{nonAF}})$

3. Kolmogorov–Smirnov statistic: $T = \sup_x |\hat{F}_{\mathrm{AF}}(x) - \hat{F}_{\mathrm{nonAF}}(x)|$

Permutation procedure. Let T_{obs} be the observed statistic computed from the original labels.

1. Fix the observed g values and randomly permute AF labels across observations.

2. For $b = 1, \ldots, B$, compute the permuted statistic T_b.

3. Compute a two-sided p-value:

$$p = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left(|T_b| \geq |T_{\mathrm{obs}}|\right)}{B + 1}.$$

4. At level α, report **reject** or **fail to reject** H_0.

Independence / repeated exams. Since the dataset may contain multiple exams per patient, we will explicitly address non-independence by providing:

- **Default (recommended):** restrict to **one exam per patient** (user may select "earliest" or "random" exam per `patient_id`).

- **Optional extension:** a **block permutation** approach at the patient level (permute group labels at the `patient_id` level), and compare results to the default approach.

Outputs provided.

- A histogram/density plot of $\{T_b\}_{b=1}^{B}$ with a vertical line at T_{obs}
- The permutation p-value and a clear reject/fail-to-reject statement
- A summary table including group sizes and descriptive statistics of g (mean, median, SD)
- (Optional) sensitivity comparison across statistics and sampling modes (per-patient vs exam-level)

4 User Inputs (Shiny Components)

Users will be able to modify the following:

1. Random seed
2. Number of permutations B (e.g., 500 / 1000 / 5000)
3. Significance level α and one-sided vs two-sided testing
4. Choice of test statistic (mean difference / median difference / KS)
5. Data handling option for repeated exams:
   - one exam per patient (earliest vs random),
   - (optional) patient-level block permutation.
6. Display options (toggle):
   - permutation distribution plot,
   - group summary table,
   - brief assumptions/caveats panel.

References

A. H. Ribeiro, M. H. Ribeiro, G. M. M. Paixão, D. M. Oliveira, P. R. Gomes, J. A. Canazart, M. P. S. Ferreira, C. R. Andersson, P. W. Macfarlane, W. J. Meira, T. B. Schön, and A. L. P. Ribeiro. CODE-15%: A Large-Scale Electrocardiogram Dataset for Deep Learning, 2021. URL `https://doi.org/10.5281/zenodo.4916206`.